SUPPLEMENTARY MATERIAL
Survival-Time Risk Beyond Low-Order Channel Models: Identifiability and Scheduling

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S1. Overview

Results of the main paper are cited by their numbers there, e.g., Lemma 1, Theorem 1, or Fig. 4; numbering with the prefix "S" refers to this document. Notation follows the main paper. Section S2 illustrates the system model, gives the design of π_L in full, and represents the risk of any policy through sampled loss patterns. Section S3 gives complete proofs and additional theory. Sections S4 and S5 report further numerical results behind the claims of the main paper, and Section S5.2 also evaluates a redesign of π_L with the true recovery curve, which the main paper does not include. Section S6 maps each claim to the script that produces it.

S2. System Model and Low-Order Design

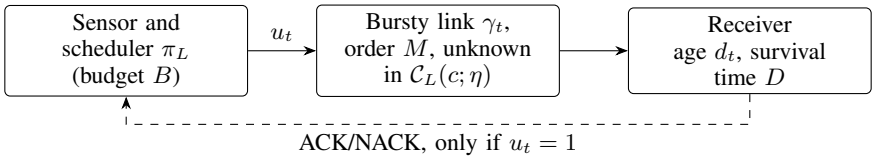


Fig. S1. System model. The latent loss process evolves in every slot; the scheduler observes it only in slots where it transmits.

Fig. S1 shows the setting of Section II. For $L = 1$, after an observation o made δ slots earlier, the design of π_L uses the prediction $q = f^{\delta-1}(q_o)$ of Section II-B, and before the first observation, denoted by $o = \varnothing$, it uses $q = c(1)$. The design solves

$$V_t(d, r, o, \delta) = \min\{V^0, V^1\}, \qquad V^0 = F\big(d{+}1, V_{t+1}(d{+}1, r, o, \delta{+}1)\big),$$
$$V^1 = q\, V_{t+1}(0, r{-}1, 1, 1) + (1{-}q)\, F\big(d{+}1, V_{t+1}(d{+}1, r{-}1, 0, 1)\big), \tag{S1}$$

where $d = d_{t-1}$ is the age at the start of slot t, $F(d', z) = 1$ if $d' \geq D$ and $F(d', z) = z$ otherwise, $V_{T+1} \equiv 0$, V^1 requires $r \geq 1$, and ties are broken toward idling. The nominal risk R_{nom} is $V_1(0, B, \varnothing, 0)$. If $B \geq T$, attempting in every slot minimizes the age pathwise on every channel, since an idle slot can only delay a reset. Where a failure is still possible, idling leads, with positive probability under the fit, to a failure that an attempt would avoid, because the fit is η-positive whenever $\mathcal{C}_L(c;\eta)$ is nonempty. Hence π_L attempts in exactly these slots, and its risk equals that of dense attempts on every channel. The same recursion with an arbitrary *recovery curve* $\Pr(\gamma_{t+\delta} = 1 \mid \gamma_t = o)$ in place of $f^{\delta-1}(q_o)$ defines the redesign π_{rec} of Section S5.2.

Proposition S1 (Sampled-pattern representation)**.** *For every deterministic policy π with budget B under ACK/NACK feedback, there is a finite set $\mathcal{H}_\pi$ of pairs $(\mathbf{s}, \mathbf{o})$ of attempt-slot tuples $\mathbf{s} = (s_1 < \cdots < s_j)$ and outcome tuples $\mathbf{o} \in \{0,1\}^j$, determined by π alone, such that for every channel $\mathcal{M}$*

$$R_{\mathcal{M}}(\pi) = \sum_{(\mathbf{s},\mathbf{o}) \in \mathcal{H}_\pi} \Pr_{\mathcal{M}}(\gamma_{s_1} = o_1, \ldots, \gamma_{s_j} = o_j). \tag{S2}$$

Proof: The scheduler's information, the age, and the remaining budget are functions of the attempted outcomes only. Enumerating the ACK/NACK histories up to the failure slot partitions $\{\tau \leq T\}$ into disjoint cylinder events on attempted slots. ∎

The risk is therefore a linear functional of gapped pattern probabilities, and the policy selects which patterns enter; if c determines all of them, it determines the risk. This sufficient condition fails in general, since with dense attempts and $D > L{+}1$, for instance, every failure pattern fixes at least D consecutive outcomes. The closed-loop decomposition Eq. (15) makes the selected patterns explicit.

S3. Proofs and Additional Theory

S3.1 Proof of Lemma 1

A stationary M-th order chain with positive context law is equivalent to a shift-consistent $(M{+}1)$-block law ν, i.e., one whose left and right M-marginals coincide and thus both equal the context law μ. Given such ν, the transition probabilities $p(w) = \nu(w1)/\mu(w)$ started from μ define a stationary chain with block law ν, and conversely. For $M = L{+}1$ the $(L{+}1)$-block law of the chain is μ itself, so compatibility means that both M-marginals of ν equal c. For a middle word $m \in \{0,1\}^L$ the four entries $\nu(a\,m\,a')$, $a, a' \in \{0,1\}$, form a 2×2 table with row sums $c(am)$ and column sums $c(ma')$; distinct middle words involve disjoint entries. Each table has one free entry $x_m = \nu(0m0)$, and the others follow as stated in the lemma. Nonnegativity yields the Fréchet interval $\max\{0, c(m0) - c(1m)\} \le x_m \le \min\{c(0m), c(m0)\}$. The constraint $\eta \le p(am) \le 1-\eta$ reads $x_m \in [\eta c(0m), (1-\eta)c(0m)]$ for $a = 0$ and $c(1m) - c(m0) + x_m \in [\eta c(1m), (1-\eta)c(1m)]$ for $a = 1$; the intersection is $[x_m^{\mathrm{lo}}, x_m^{\mathrm{hi}}]$ and implies nonnegativity. Since the tables are independent, the class is the product of these intervals, and the map is affine.

The entropy rate of the chain is $\bar{H} = H(\gamma_{M+1} \mid \gamma_{1:M}) = H(\gamma_{L+2} \mid \gamma_1, \gamma_{2:L+1}) \le H(\gamma_{L+2} \mid \gamma_{2:L+1})$, and the right side depends only on c. Equality holds iff γ_1 and γ_{L+2} are conditionally independent given every middle word, i.e., $\nu(a\,m\,a') = c(am)c(ma')/c(m)$, which gives x_m^{nom}; this is the fitted L-th order model, whose transition $\Pr(a' \mid m) = c(ma')/c(m)$ ignores a. Since $c(m1)/c(m) = [c(0m)p(0m) + c(1m)p(1m)]/c(m)$ for every compatible channel, the fitted model is η-positive, and hence the unique maximizer, whenever the class is nonempty. In the running example, the class is parameterized by (ρ, b) through $p(00) = 1-\rho$, $p(01) = b$, $p(10) = 1 - \frac{3}{2}(1-\rho)$, and $p(11) = 1 - \frac{b}{9}$.

S3.2 Proof of Theorem 1 and Eq. (12)

The schedule is open-loop, so its attempts do not depend on failure, and the age process can be continued past D; the indicator $I_t = \mathbf{1}\{d_t = D\}$ marks the slots in which the age equals D. The age increases by one per slot until it is reset, so it cannot jump over D, and $\{\tau \le T\} = \{\sum_{t\le T} I_t \ge 1\}$. Moreover, $I_t = 1$ iff the last reset before t occurred at $t - D \in \mathcal{S}_0$ and no attempt in $(t-D, t]$ succeeded, i.e., iff $\bar{A}_{t-D}$ occurs; hence $\mathbb{E}\sum_t I_t = \Lambda_{\mathcal{S}}$, and the union bound gives $R \le \Lambda_{\mathcal{S}}$. This step uses neither the Markov property nor stationarity.

For two hits $t < t'$, the reset at $t' - D$ comes after t, since for $t' - D \le t$ the age at t would be $t - (t' - D) < D$ or 0. Hence $t' - D > t$, so I_t is measurable with respect to $\mathcal{F}_{t'-D-1} = \sigma(S_1, \gamma_{1:t'-D-1})$, and $t' - D \in \mathcal{S}$ because $t' - D > t \ge D \ge 1$. By the Markov property, $\mathbb{E}[I_{t'} \mid \mathcal{F}_{t'-D-1}] = \Pr(A_{t'-D} \mid S_{t'-D}) \le \kappa_{\mathcal{S}}$. For fixed $t = s + D$ there are at most $J_{\mathcal{S}}$ admissible t', so $\sum_{t<t'} \mathbb{E}[I_t I_{t'}] \le J_{\mathcal{S}}\kappa_{\mathcal{S}}\Lambda_{\mathcal{S}}$, and the second Bonferroni inequality $\Pr(\cup E_i) \ge \sum \Pr(E_i) - \sum_{i<j} \Pr(E_i E_j)$ yields the lower bound.

For dense attempts, $A_0 = \{0^D\}$ and $A_s = \{\gamma_s = 1\} \cap \{0^D \text{ on } (s, s+D]\}$ for $s = 1, \dots, n$. Since $\Pr(0^{M+1}) = x_{0^L} = \rho\, c(0^M)$ and every further loss after M losses has probability ρ, $\Pr(0^D) = c(0^M)\rho^k$ and, by stationarity, $\Pr(10^D) = \Pr(0^D) - \Pr(0^{D+1}) = c(0^M)(1-\rho)\rho^k$. Given $S_s = w$, A_s requires a success followed by D losses, so $\Pr(A_s \mid w) \le \rho^k \max_{v: v_M = 1} \Pr(0^M \mid v)$, and $J = (T - 2D)^+ \le n$; this gives $J\kappa \le \theta_{\mathcal{M}}\Lambda$. In the running example ($M = 2$), the contexts ending in a success are 01 and 11, and $\Pr(00 \mid v) = (1 - p(v))(1 - p(10))$ with $1 - p(10) = \frac{3}{2}(1-\rho)$, $1 - p(01) = 1 - b$, and $1 - p(11) = \frac{b}{9}$; dividing by $c(00)(1-\rho) = 0.12(1-\rho)$ gives $\theta_{\mathcal{M}} = 12.5 \max\{1 - b, \frac{b}{9}\} \le 6.9$.

S3.3 Proof of Theorem 2

Throughout, $k \ge 1$ and $n = T - D \ge 0$, and $0 < \rho_{\mathrm{lo}} \le \rho_{\mathrm{hi}} < 1$ because every compatible channel has $\eta \le \rho \le 1 - \eta$. With $B \ge T$, the risks $R^{\pm}$ and R_{nom} are those of dense attempts.

Monotonicity and unimodality. With $\varphi(\rho) = \rho^k[1{+}n(1{-}\rho)]$, $\varphi'(\rho) = \rho^{k-1}[k(n{+}1){-}(k{+}1)n\rho]$. The bracket is nonincreasing in ρ and equals $k - n$ at $\rho = 1$. If $n \le k$, it is positive on $(0,1)$, so φ is increasing and ρ_{hi} maximizes $\Lambda = c(0^M)\varphi(\rho)$, which proves (i). If $n > k$, it vanishes at $\rho^\star = k(n+1)/[(k+1)n] \in (k/(k+1), 1)$, so φ increases on $(0, \rho^\star]$ and decreases on $[\rho^\star, 1)$. On $[\rho_{\mathrm{lo}}, \rho_{\mathrm{hi}}]$, the maximum is therefore attained at the point closest to $\rho^\star$, i.e., at $\rho_w = \min\{\max\{\rho^\star, \rho_{\mathrm{lo}}\}, \rho_{\mathrm{hi}}\}$, and the minimum at an endpoint, since a unimodal function has no interior minimum. This proves (ii).

Most persistent channel. By (i) and (ii), ρ_{hi} maximizes Λ for a given n if and only if $n \le k$ or $\rho^\star(n) \ge \rho_{\mathrm{hi}}$. For $n > k$, $\rho^\star(n) = \frac{k}{k+1}(1 + \frac{1}{n})$ decreases strictly in n toward $k/(k+1)$ without attaining it. Hence $\rho^\star(n) \ge \rho_{\mathrm{hi}}$ for all $n > k$ if and only if $k/(k+1) \ge \rho_{\mathrm{hi}}$, i.e., $k \ge \rho_{\mathrm{hi}}/(1 - \rho_{\mathrm{hi}})$. As $T \ge D$ ranges over all $n \ge 0$, this proves (iii).

Growth with D. For $\rho' < \rho_{\mathrm{hi}}$, let $x = 1 - \rho_{\mathrm{hi}}$ and $y = 1 - \rho'$, so $0 < x < y$. Then $\varphi(\rho_{\mathrm{hi}})/\varphi(\rho') = (\rho_{\mathrm{hi}}/\rho')^k(1{+}nx)/(1{+}ny)$, and $y(1+nx) - x(1+ny) = y - x > 0$ gives $(1+nx)/(1+ny) > x/y$, which is the inequality in (iv). For the ratios, let C_{hi} and C_{lo} be compatible channels with $\rho = \rho_{\mathrm{hi}}$ and $\rho = \rho_{\mathrm{lo}}$, and let C_{nom} be the fitted model, which is compatible whenever the class is nonempty. By Theorem 1 and $J\kappa \le \theta_{\mathcal{M}}\Lambda$ from the previous subsection, every compatible channel satisfies $\Lambda(1 - \theta_{\mathcal{M}}\Lambda) \le R \le \Lambda$ with $\Lambda = c(0^M)\varphi(\rho)$. Since $R^+ \ge R(C_{\mathrm{hi}})$ and $R_{\mathrm{nom}} = R(C_{\mathrm{nom}})$,

$$\frac{R^+}{R_{\mathrm{nom}}} \ge (1 - \theta_{\mathrm{hi}}\Lambda_{\mathrm{hi}})\frac{\varphi(\rho_{\mathrm{hi}})}{\varphi(\rho_{\mathrm{nom}})} \ge (1 - \theta_{\mathrm{hi}}\Lambda_{\mathrm{hi}})\Big(\frac{\rho_{\mathrm{hi}}}{\rho_{\mathrm{nom}}}\Big)^k \frac{1 - \rho_{\mathrm{hi}}}{1 - \rho_{\mathrm{nom}}},$$

where θ_{hi} and Λ_{hi} refer to C_{hi}. Since $R^- \le R(C_{\mathrm{lo}})$, the same bound holds for R^+/R^- with ρ_{lo} in place of ρ_{nom}. Apart from the factor $1-\theta_{\mathrm{hi}}\Lambda_{\mathrm{hi}}$, the bound does not depend on n, and its exponent is $k = D - M$. Each additional slot of D therefore multiplies the bounds on R^+/R_{nom} and R^+/R^- by $\rho_{\mathrm{hi}}/\rho_{\mathrm{nom}}$ and $\rho_{\mathrm{hi}}/\rho_{\mathrm{lo}}$, respectively, where $\theta_{\mathrm{hi}}\Lambda_{\mathrm{hi}}$ stays small, i.e., in the rare-failure regime of (iv).

S3.4 Consequences of Theorem 2

Fitted model. The fitted model maximizes Λ only if $\rho_{\mathrm{nom}} = \rho_w$; in the running example this happens, e.g., for $k = 1$, $n = 5$ ($D = 3$, $T = 8$), where $\rho^\star = \frac{6}{10} = \rho_{\mathrm{nom}}$, although even there the exact risk is largest at $b = 0.95$. Generically, the fitted model is not the worst case.

Required precision. The sensitivity $\partial \log\Lambda/\partial\rho = k/\rho - n/[1+n(1-\rho)]$ equals 14.3 at $\rho_{\mathrm{nom}} = 0.6$ for $T = 60$ and $D = 12$; an error of $\Delta\rho$ multiplies the risk by about $e^{14.3\,\Delta\rho}$, so a factor-two accuracy requires $|\Delta\rho| \lesssim 0.049$, and the required precision grows linearly in D. In the running example, $\rho = \Pr(0 \mid 00)$ is estimated from about $c(00)N = 0.12N$ occurrences of the context 00 in N slots, so an accuracy of 0.049 at two standard errors requires $N \approx 3.4 \times 10^3$ slots, where the dependence between occurrences is neglected. With a known order, this sample size is moderate, but it grows roughly as $D^2/c(0^M)$.

S3.5 Decay rates and decimated persistences

Proposition S2 (Decay exponents)**.** *With Q_g and σ_g as in Section IV-A, $\sigma_1 = \rho$, $\sigma_g \ge \rho^g$, and Q_g is entrywise positive for $g \ge M$. If $\log(1+T-D) = o(D)$ and either $g = 1$ or $g \ge M$, then Eq. (14) holds.*

Proof: For $s \in \mathcal{S}_g$, define $\alpha_s(w) = \Pr(\gamma_s = 1, \gamma_{s+g} = 0, S_{s+g+1} = w)$ on loss-ending contexts; then $\Pr(A_s) = \alpha_s^\top Q_g^{D_g-1}\mathbf{1}$, and similarly for A_0. By stationarity α_s does not depend on s, and $\Lambda_{\mathcal{S}_g}$ has at most $n/g+1$ such terms. For $g \ge M$, Q_g is entrywise positive, so the Perron–Frobenius theorem gives $Q_g^j = \sigma_g^j(\xi\zeta^\top + o(1))$ with positive ξ, ζ; hence $\Pr(A_s) = \Theta(\sigma_g^{D_g})$ and $\log\Lambda_{\mathcal{S}_g} = D_g \log\sigma_g + O(\log n) + O(1)$. Since every context has η-positive success probability, Q_g is strictly substochastic and $\sigma_g < 1$. The second-order term satisfies $J\kappa \le (n/g)\max_w e_w^\top Q_g^{D_g-1}\mathbf{1} = O(n\sigma_g^{D_g}) \to 0$ when $\log n = o(D)$, so $\log R = \log\Lambda + o(1)$ and $\frac{1}{D}\log R \to \frac{1}{g}\log\sigma_g$. For $g = 1$, Q_1 is not primitive, but Eq. (12) gives Λ in closed form, $\frac{1}{D}\log\Lambda \to \log\rho$, and $\Lambda \to 0$, so that $J\kappa \le \theta_{\mathcal{M}}\Lambda \to 0$ as in Section III-B. In the directed graph of the positive entries of Q_1, the only cycle is the self-loop at 0^M, so $\sigma_1 = \rho$. Finally, $\sigma_g \ge Q_g(0^M, 0^M) \ge \rho^g$, since the spectral radius of a nonnegative matrix dominates every diagonal entry. For $g \ge M$, all $g-1$ intermediate outcomes are free, so every loss-ending context is reached with η-positive probability, and Q_g is entrywise positive, as used above. ∎

For $M = 2$, $g = 2$, and contexts ordered as $(00, 10)$,

$$Q_2 = \begin{pmatrix} \rho^2 & (1-\rho)(1-b) \\ (1-p(10))\rho & p(10)(1-b) \end{pmatrix}, \tag{S3}$$

whose second column holds the relapse terms, i.e., the loss–success–loss transitions. For M-th order chains and $g \ge M$, the run persistences $\rho_j^{(g)}$ of Proposition 1 converge to σ_g as $j \to \infty$, because Q_g is entrywise positive and $\Pr^{(g)}(0^j) = \sum_w \mu(w)\, e_w^\top Q_g^{j-1}\mathbf{1}$, with the sum over loss-ending contexts w. For C_1 and C_2, $\sigma_2 = 0.238$ and 0.433, respectively. The 2-decimated persistences of C_1 start at $\rho_1^{(2)} = 0.284$ and remain between 0.2326 and 0.284 for $j \le 60$, and those of C_2 start at 0.44 and remain between 0.4293 and 0.44; at $j = 60$, both match σ_2 to eight digits. Hence those of C_1 lie below those of C_2 at every depth. At $n = 48$, exact risks give log-ratio slopes of 0.1823 for dense and 0.2986 for two-slot periodic attempts per unit of D, matching $\log(\rho(C_1)/\rho(C_2)) = 0.1823$ and $\frac{1}{2}\log(\sigma_2(C_2)/\sigma_2(C_1)) = 0.2986$, and the reversal holds for every even D from 4 to 40. In general, where $g \ge M$ and $\log(1+n) = o(D)$, the log-risk ratios of two channels $\mathcal{M}$ and $\mathcal{M}'$ under dense and g-spaced attempts have the asymptotic slopes $\log(\rho(\mathcal{M})/\rho(\mathcal{M}'))$ and $\frac{1}{g}\log(\sigma_g(\mathcal{M})/\sigma_g(\mathcal{M}'))$ per unit of D, so the two schedules rank the channels oppositely for large D whenever $\rho(\mathcal{M}) > \rho(\mathcal{M}')$ but $\sigma_g(\mathcal{M}) < \sigma_g(\mathcal{M}')$.

S3.6 Closed-loop decomposition

Proposition S3 (Closed-loop decomposition)**.** *For every deterministic policy π under ACK/NACK feedback, $R(\pi) = \Lambda_\pi$ with Λ_π as in Eq. (15).*

Proof: After a success at s, the scheduler state is a function of the observation history h up to s; for π_L it is $(d, o, \delta) = (0, 1, 1)$ together with the remaining budget r. As long as the attempts fail, the observations are NACKs, so the decisions in $(s, s+D]$ are a deterministic function of (s, h), the retry pattern $\mathcal{R}_\pi(s, h)$. If $\tau \le T$, the last success before τ, or the start $s = 0$, is unique, precedes any violation, and is followed by losses at all attempts of $\mathcal{R}_\pi(s, h)$; moreover, $\tau = s + D$. Conversely, a success at s with $s + D \le T$ before any violation, followed by such losses, gives $\tau = s + D$. The events $E_{s,h,w} \cap \{\gamma_j = 0\ \forall j \in \mathcal{R}_\pi(s, h)\}$ therefore partition $\{\tau \le T\}$. Since $E_{s,h,w}$ is determined by S_1 and $\gamma_{1:s}$ and fixes $S_{s+1} = w$, the Markov property gives $\Pr(E_{s,h,w}, \gamma_{\mathcal{R}_\pi(s,h)} = 0) = \Pr(E_{s,h,w})\Pr(\gamma_{\mathcal{R}_\pi(s,h)} = 0 \mid S_{s+1} = w)$, and summing over the partition yields $R(\pi) = \Lambda_\pi$. For the start ($s = 0$), h is empty, $r = B$, and $S_1 \sim \mu$. ∎

Unlike $\Lambda_{\mathcal{S}}$ in Theorem 1, Λ_π weights each retry pattern by the probability of a success before any violation, which depends on π and follows from the same forward recursion as R. The identity therefore shows which loss patterns enter the risk rather than giving a closed form. If π is continued after a violation, e.g., by attempting until the next success, and the later successes are also counted, the sum becomes an upper bound on $R(\pi)$, as $\Lambda_{\mathcal{S}}$ is for open-loop schedules. For π_L ($T = 60$, $B = 15$, $D = 12$) on a 9×5 grid of the class, the computed Λ_π matches R to within a relative error of 10^{-14}.

S3.7 No schedule-independent burstiness index

Corollary S1. *In the running example, no function ψ on $\mathcal{C}_L(c;\eta)$, such as the burst-length tail, a lag-g correlation with $g > L$, ρ, or a Gilbert–Elliott parameter, satisfies $R_{\mathcal{M}}(\pi_L) < R_{\mathcal{M}'}(\pi_L) \Rightarrow \psi(\mathcal{M}) < \psi(\mathcal{M}')$ for every (T, D, B).*

Proof: Item 1) of Proposition 2 forces $\psi(C_2) < \psi(C_1)$, and item 3) forces $\psi(C_1) < \psi(C_2)$. ■

S3.8 Order mismatch $M \geq L + 2$

Remark S1. *For $M \geq L + 2$ the compatible class is a general polytope, and $y = \Pr(0^M)$ is no longer fixed by c. With $x = \Pr(0^{M+1})$, Eq. (12) generalizes to $\Lambda = y\,\varphi(x/y)$, which is homogeneous of degree one in (y, x). For fixed $\rho = x/y$, Λ is proportional to y, so the maximum over the class lies where a ray of slope ρ leaves the two-dimensional projection of the polytope onto (y, x), i.e., on its boundary; the projection follows from linear programs. For the running example with $M = 3$ instead of $M = 2$ ($T = 60$, $D = 12$), $y \in [0.044, 0.114]$ with nominal value 0.072, and a grid search over the projected polygon gives $\Lambda_{\max} = 0.254$, against 0.253 for $M = 2$, at $y = 0.112$ and $x/y = 0.928$. Since the $M = 3$ class contains the $M = 2$ class, whose dense worst case is 0.242, the $M = 3$ dense worst case lies in $[0.242, 0.254]$. The closed-loop worst case over the polytope remains open.*

S4. Additional Results for the Running Example

S4.1 Dense attempts

Table S1 compares the worst-case persistence of Theorem 2 with exact maximizers over a grid of 1201 values of ρ at $b = 0.7$. The maximizer of Λ matches ρ_w to grid resolution, and the maximizer of R deviates slightly where R saturates. The two-sided bound of Theorem 1 is violated by at most 1.4×10^{-16} for the (T, D) of Table S1 with $b \in \{0.45, 0.7, 0.95\}$. For random channels of the class under dense, periodic, and irregular open-loop schedules ($T = 60$, $D = 12$), it is never violated, and its gap $(\Lambda_{\mathcal{S}} - R)/R$ stays below 16%. In the phase diagram of Fig. S2, the worst channel is interior for long missions, and the cap $\rho_{\text{hi}} = 0.95$ binds for every mission length once $D \geq D^\star = 21$. The right panel shows that the risk ratios R^+/R_{nom} and R^+/R^- grow geometrically in D, as predicted by Λ, except at small D, where R^+ saturates. Table S2 repeats the comparison for a third-order class ($L = 2$, $M = 3$) whose fitted model is the second-order chain with $(p(00), p(01), p(10), p(11)) = (0.35, 0.85, 0.55, 0.92)$, a four-dimensional box with $\rho \in [0.488, 0.950]$ and $\rho_{\text{nom}} = 0.650$. Its column "spread" is the largest relative change of the dense risk when the three relapse coordinates are redrawn at fixed ρ; it is at most 2.7%.

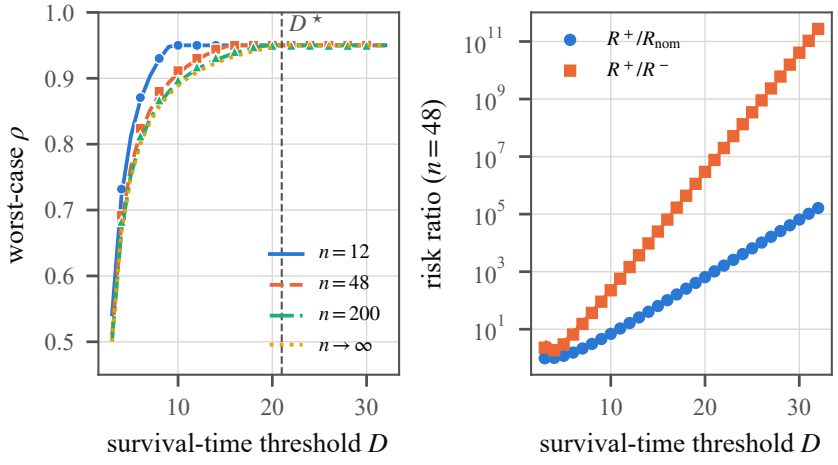


Fig. S2. Worst-case persistence ρ_w versus D for several $n = T - D$ (left) and risk ratios for $n = 48$ (right). Lines show Theorem 2 and the predictions from Λ, and markers show exact risk maximizers and exact ratios, with ρ varied at $b = 0.7$. The cap $\rho_{\text{hi}} = 0.95$ is active for all n once $D \geq D^\star = 21$.

TABLE S1
DENSE ATTEMPTS OVER ρ AT $b = 0.7$ IN THE RUNNING EXAMPLE

T	D	ρ_w (Thm.)	argmax Λ	argmax R	R^+	R^-	R_{nom}	R^+/R_{nom}
6	6	0.950	0.950	0.950	9.8×10^{-2}	2.2×10^{-3}	1.6×10^{-2}	6.3
12	6	0.933	0.933	0.933	1.3×10^{-1}	1.0×10^{-2}	5.3×10^{-2}	2.4
30	6	0.833	0.833	0.840	2.7×10^{-1}	3.5×10^{-2}	1.6×10^{-1}	1.7
60	6	0.815	0.815	0.822	4.7×10^{-1}	7.4×10^{-2}	3.1×10^{-1}	1.5
120	6	0.807	0.807	0.815	7.2×10^{-1}	1.5×10^{-1}	5.3×10^{-1}	1.4
30	10	0.933	0.933	0.935	1.6×10^{-1}	5.4×10^{-4}	1.8×10^{-2}	8.8
60	10	0.907	0.907	0.910	2.9×10^{-1}	1.3×10^{-3}	4.2×10^{-2}	6.8
200	10	0.894	0.894	0.898	6.8×10^{-1}	4.7×10^{-3}	1.4×10^{-1}	4.7
100	15	0.939	0.939	0.942	3.0×10^{-1}	1.4×10^{-5}	5.5×10^{-3}	55
300	15	0.932	0.932	0.935	6.6×10^{-1}	4.7×10^{-5}	1.8×10^{-2}	37
100	24	0.950	0.950	0.950	1.8×10^{-1}	1.5×10^{-9}	5.0×10^{-5}	$3.6k$
400	24	0.950	0.950	0.950	5.6×10^{-1}	7.4×10^{-9}	2.4×10^{-4}	$2.4k$

TABLE S2
DENSE ATTEMPTS OVER THE THIRD-ORDER CLASS ($L = 2$, $M = 3$)

T	D	ρ_w (Thm.)	argmax R	R^+	R^-	R_{nom}	spread
8	8	0.950	0.950	4.6×10^{-2}	1.7×10^{-3}	7.0×10^{-3}	0.0%
60	8	0.849	0.847	2.2×10^{-1}	4.5×10^{-2}	1.3×10^{-1}	2.7%
200	8	0.838	0.842	5.7×10^{-1}	1.5×10^{-1}	3.9×10^{-1}	2.4%
100	12	0.910	0.913	2.1×10^{-1}	4.4×10^{-3}	3.9×10^{-2}	1.6%
400	12	0.902	0.899	6.2×10^{-1}	1.9×10^{-2}	1.6×10^{-1}	1.5%
150	30	0.950	0.950	1.0×10^{-1}	1.5×10^{-8}	2.3×10^{-5}	0.5%

S4.2 Budgeted attempts, certificates, and the positivity bound

Table S3 is the closed-loop sweep over the budget summarized in Fig. 3 ($T = 60$, $D = 12$), evaluated on a grid of 47×21 channels in (ρ, b) with resolution 0.013 in ρ. As B decreases, the worst channel moves toward less persistent channels with frequent relapses ($b = 0.45$), and the best channel, at ρ_{lo}, has $b = 0.45$ for $B = 60$ and $b = 0.95$ for all smaller budgets. At $\rho = \frac{11}{30}$, switching b from 0.95 to 0.45 leaves the dense risk nearly unchanged but raises the risk of periodic attempts every four slots by $3.88\times$, every two slots by $34.6\times$, and that of π_L with $B = 15$ by $9.05\times$; at $\rho = 0.95$ the same switch changes the dense, every-four-slots, and π_L risks by at most 19%.

Table S4 lists the certificates discussed in Section V ($T = 60$). For dense attempts, its worst cases are taken over the grid of Table S3 and a search over 2001 values of ρ at $b \in \{0.45, 0.95\}$, which raises R^+ at $D = 12$ from 0.241 to 0.242. The interval relaxation is the rectangular robust MDP of Section V-A; the time-invariant corner instead fixes the worst corner of the projected intervals for the whole mission and starts in the stationary law of that corner. Parameter lifting [1] also relaxes the coupling of the box, but its bounds tighten as the parameter space is subdivided.

Table S5 varies the positivity bound η ($T = 60$, $D = 12$), which does not affect $R_{\mathrm{nom}} = 1.46 \times 10^{-2}$, and maximizes over ρ at the midpoint of the range of b; at $\eta = 0.05$, this gives $R^+ = 0.240$ instead of 0.242. For $\eta \le 0.05$, the dense worst case is interior and barely moves. For $\eta = 0.1$, $\rho_{\mathrm{hi}} = 0.9$ lies below $\rho^\star = 0.928$, so the worst case moves to ρ_{hi}, as Theorem 2(ii) predicts.

TABLE S3
CLOSED-LOOP RISK OF π_L OVER THE COMPATIBLE CLASS

B	R_{nom}	R^+	at (ρ, b)	R^-	R^+/R_{nom}
60	1.5×10^{-2}	0.241	(0.925, 0.95)	1.7×10^{-4}	16.5
30	2.2×10^{-2}	0.254	(0.925, 0.45)	4.7×10^{-4}	11.4
20	3.7×10^{-2}	0.269	(0.912, 0.45)	1.7×10^{-3}	7.3
15	6.0×10^{-2}	0.286	(0.899, 0.45)	6.7×10^{-3}	4.8
12	9.3×10^{-2}	0.303	(0.887, 0.45)	2.2×10^{-2}	3.3

TABLE S4
RISK CERTIFICATES FOR THE RUNNING EXAMPLE

Task	R_{nom}	R^+ (grid)	box $\Lambda_{\max}$	interval relax.	time-inv. corner
dense, $D = 12$	1.5×10^{-2}	0.242	0.253	0.973	0.967
dense, $D = 20$	2.1×10^{-4}	0.142	0.143	0.795	0.782
$B = 15$, $D = 12$	6.0×10^{-2}	0.286	–	0.985	0.982

TABLE S5
SENSITIVITY OF THE DENSE WORST CASE TO η

η	$[\rho_{\mathrm{lo}}, \rho_{\mathrm{hi}}]$	worst ρ	R^+	R^-	R^+/R_{nom}
0.01	$[0.340, 0.990]$	0.931	0.239	8.1×10^{-5}	16.3
0.02	$[0.347, 0.980]$	0.931	0.239	9.7×10^{-5}	16.4
0.05	$[0.367, 0.950]$	0.931	0.240	1.7×10^{-4}	16.5
0.1	$[0.400, 0.900]$	0.900	0.230	3.7×10^{-4}	15.8

S4.3 Exact certificate of Proposition 2

The risks of Proposition 2 and the design of π_L behind them were computed in exact rational arithmetic with Python's `fractions.Fraction`, which gives the fractions listed below.

- dense attempts, $R(C_1) \approx 9.096922\times10^{-4}$:
  `28764046942402247011864795479079374618670119590699773794325455937275452419409017556257663661031306070096308800072783045878393774943/31619537076459609040430879785135516613081563264131546020507812500000000000

projection is the second-order Markov chain with the same 3-block distribution; ρ and b are its coordinates, and "proj." is its risk. The last column gives, for $B = 15$, the risks of the redesign π_{rec} of Section S5.2 and of a genie that observes the hidden state in every slot. For $e_G = 0$, a loss reveals the bad state, so the persistence is constant at every burst depth and the dense risks agree; relapses created by successes in the bad state raise the budgeted risk by up to $2.1\times$.

TABLE S7
GILBERT–ELLIOTT CHANNELS SHARING THE GILBERT FIT OF THE RUNNING EXAMPLE

(e_G, e_B)	memory	ρ	b	ρ_∞	dense	proj.	π_L, B=15	proj.	π_{rec} / genie
$(0, 1)$	0.50	0.600	0.900	0.600	1.46×10^{-2}	1.46×10^{-2}	5.97×10^{-2}	5.97×10^{-2}	6.0×10^{-2} / 2.6×10^{-2}
$(0, 0.8)$	0.67	0.600	0.733	0.600	1.46×10^{-2}	1.46×10^{-2}	8.25×10^{-2}	8.20×10^{-2}	7.6×10^{-2} / 5.9×10^{-2}
$(0, 0.7)$	0.80	0.600	0.600	0.600	1.46×10^{-2}	1.46×10^{-2}	1.06×10^{-1}	1.02×10^{-1}	8.4×10^{-2} / 7.7×10^{-2}
$(0, 0.65)$	0.89	0.600	0.511	0.600	1.46×10^{-2}	1.46×10^{-2}	1.27×10^{-1}	1.17×10^{-1}	9.3×10^{-2} / 8.8×10^{-2}
$(0.01, 0.9)$	0.60	0.618	0.826	0.618	1.88×10^{-2}	1.88×10^{-2}	7.89×10^{-2}	7.77×10^{-2}	7.8×10^{-2} / 5.2×10^{-2}

S5.2 Designing with the recovery curve

The redesign π_{rec} runs Eq. (S1) with the channel's true recovery curve $\Pr(\gamma_{t+\delta} = 1 \mid \gamma_t = o)$ instead of the Gilbert prediction $f^{\delta-1}(q_o)$. After a loss, this curve equals $1 - \rho_1^{(\delta)}$ with $\rho_1^{(\delta)}$ as in Section IV-A; it can be estimated from a trace or learned from ACK/NACK feedback [2]. Table S8 uses $T = 60$, $D = 12$, and $B = 15$ and compares both policies with a genie that observes the channel context in every slot and thus lower-bounds every policy under ACK/NACK feedback. The redesign lowers the risk on C_2 from 6.04×10^{-2} to 2.68×10^{-2} but raises it on C_1 from 1.30×10^{-2} to 1.67×10^{-2}, and for the Gilbert–Elliott channel with $e_B = 0.65$ in Table S7, it lowers the risk from 0.127 to 0.093, 6% above the state-observing genie. On a 9×5 grid of the class, it wins on every channel with $b \le 0.575$, where relapses are frequent, but loses on 23 of 45, all with $b \ge 0.7$, by up to $1.47\times$. A designer who knows the channel can evaluate both policies exactly and keep the better one, which over the grid improves on π_L by up to $2.25\times$ and has 1.0 to 5.3 times the risk of the genie. Since π_{rec} conditions only on the last observation, it is not the optimal policy under ACK/NACK feedback for a second-order channel. The optimal policy, which requires a dynamic program over beliefs, is not computed here; its risk lies between that of the genie and that of the better of π_L and π_{rec}.

On the measured links of Section S5.5, the redesign with the empirical recovery curve improves on π_L by a median factor of 1.015 and at most 1.27, but it is worse on 52 of the 171 links. These gains are in-sample, because the empirical curve is estimated from the same tests on which both policies are evaluated.

TABLE S8
DESIGNING WITH THE TRUE RECOVERY CURVE

channel	ρ	b	$R(\pi_L)$	$R(\pi_{\mathrm{rec}})$	genie
nominal	0.600	0.90	5.97×10^{-2}	5.97×10^{-2}	4.70×10^{-2}
C_1	0.440	0.95	1.30×10^{-2}	1.67×10^{-2}	6.81×10^{-3}
C_2	0.367	0.45	6.04×10^{-2}	2.68×10^{-2}	6.13×10^{-3}

S5.3 Ambiguity across Gilbert fits

Table S9 contains the data of Fig. 4 ($T = 60$, $\eta = 10^{-4}$) together with the worst channels, the mean burst length ℓ of the fit, and the survival time D, chosen as the smallest value at which the Gilbert fit certifies $R_{\mathrm{nom}} \le 10^{-4}$ with dense attempts.

S5.4 Fading channels

Generation. Rayleigh fading with Clarke autocorrelation $J_0(2\pi f_D|t-t'|)$ between slots t and t' is generated by the inverse-DFT method of [3] on blocks of 2^{23} slots. A packet is lost if the unit-mean channel power gain $|h|^2$ falls below $-\ln(1-\bar{e})$, which gives the loss rate $\bar{e}$. For composite fading, $|h|^2$ is multiplied by $10^{X_t/10}$, where X_t is a Gauss–Markov process with standard deviation 6 dB and correlation time 200 slots, and the threshold is the empirical $\bar{e}$-quantile of the received power. Four independent blocks are used per channel (3.4×10^7 slots).

Evaluation. Missions of $T = 60$ slots start every third slot; the trace-driven risk is the fraction of failing missions. Order-L fits use $(L+1)$-block frequencies of the same trace with a pseudo-count of 0.5. Since the risk at a given D can differ by more than an order of magnitude across channels, each channel is evaluated at three survival times with trace-driven dense risks between 10^{-3} and 10^{-1}. Table S10 gives all configurations, and Table I shows those discussed in the main text. The ratios in Table S10 are fitted-to-trace risk ratios of order-L Markov fits; g=2 denotes periodic attempts every second slot, and the

TABLE S9
AMBIGUITY ACROSS GILBERT FITS

loss	ℓ	D	R_{nom}	R^+/R_{nom}	worst ρ	R_{nom}, B=15	R^+/R_{nom}, B=15	worst (ρ, b), B=15
0.5%	1.25	6	7.1×10^{-5}	68	0.816	2.8×10^{-3}	4.1	(0.625, 0.27)
0.5%	1.5	8	8.2×10^{-5}	69	0.875	7.8×10^{-4}	10	(0.325, 0.03)
0.5%	2	12	6.1×10^{-5}	86	0.925	2.2×10^{-4}	30	(0.900, 0.28)
0.5%	2.5	15	7.4×10^{-5}	67	0.950	1.7×10^{-4}	40	(0.950, 0.05)
0.5%	3	18	7.6×10^{-5}	61	0.962	1.3×10^{-4}	39	(0.937, 0.06)
0.5%	5	28	8.9×10^{-5}	46	0.994	1.1×10^{-4}	41	(0.981, 0.10)
1%	1.25	7	2.8×10^{-5}	285	0.850	1.2×10^{-3}	14	(0.750, 0.26)
1%	1.5	9	5.3×10^{-5}	182	0.891	5.7×10^{-4}	25	(0.725, 0.01)
1%	2	13	6.0×10^{-5}	162	0.933	2.1×10^{-4}	61	(0.900, 0.02)
1%	2.5	16	8.7×10^{-5}	107	0.955	2.0×10^{-4}	53	(0.933, 0.27)
1%	3	19	9.9×10^{-5}	89	0.967	1.8×10^{-4}	59	(0.950, 0.03)
1%	5	31	8.4×10^{-5}	95	0.9999	1.0×10^{-4}	81	(0.987, 0.05)
2%	1.25	7	5.6×10^{-5}	285	0.850	3.0×10^{-3}	12	(0.750, 0.25)
2%	1.5	10	3.5×10^{-5}	493	0.908	4.3×10^{-4}	62	(0.850, 0.26)
2%	2	14	5.9×10^{-5}	305	0.942	2.3×10^{-4}	95	(0.925, 0.26)
2%	2.5	18	6.0×10^{-5}	278	0.961	1.5×10^{-4}	131	(0.950, 0.26)
2%	3	21	8.4×10^{-5}	193	0.975	1.6×10^{-4}	124	(0.950, 0.01)
2%	5	34	7.9×10^{-5}	203	0.9999	9.9×10^{-5}	171	(0.987, 0.02)
5%	1.25	8	2.7×10^{-5}	$1.2k$	0.875	2.9×10^{-3}	22	$(0.800, 2.3\times10^{-3})$
5%	1.5	10	8.7×10^{-5}	490	0.908	1.6×10^{-3}	42	(0.850, 0.25)
5%	2	15	7.2×10^{-5}	579	0.950	3.7×10^{-4}	156	$(0.925, 3.7\times10^{-3})$
5%	2.5	19	8.8×10^{-5}	451	0.967	2.7×10^{-4}	185	$(0.933, 4.6\times10^{-3})$
5%	3	23	8.9×10^{-5}	426	0.979	2.0×10^{-4}	249	$(0.987, 5.6\times10^{-3})$
5%	5	37	9.1×10^{-5}	440	0.9999	1.3×10^{-4}	341	$(0.987, 9.4\times10^{-3})$
10%	1.25	8	5.5×10^{-5}	$1.2k$	0.875	8.7×10^{-3}	14	$(0.825, 1.0\times10^{-3})$
10%	1.5	11	5.7×10^{-5}	$1.3k$	0.916	1.9×10^{-3}	69	$(0.875, 1.3\times10^{-3})$
10%	2	16	7.0×10^{-5}	$1.1k$	0.958	5.7×10^{-4}	170	$(1.0\times10^{-4}, 1.7\times10^{-3})$
10%	2.5	21	6.1×10^{-5}	$1.2k$	0.972	2.8×10^{-4}	370	$(0.983, 2.2\times10^{-3})$
10%	3	25	7.5×10^{-5}	954	0.987	2.4×10^{-4}	372	$(0.975, 2.6\times10^{-3})$
10%	5	40	8.3×10^{-5}	961	0.9999	1.4×10^{-4}	574	$(0.994, 4.4\times10^{-3})$
20%	1.25	9	2.1×10^{-5}	$5.3k$	0.891	1.1×10^{-2}	14	(0.825, 0.9999)
20%	1.5	12	3.7×10^{-5}	$3.7k$	0.933	3.6×10^{-3}	48	(0.900, 0.25)
20%	2	17	6.9×10^{-5}	$2.1k$	0.958	1.3×10^{-3}	172	$(0.950, 7.0\times10^{-4})$
20%	2.5	22	7.1×10^{-5}	$2.0k$	0.978	6.6×10^{-4}	240	$(0.950, 9.0\times10^{-4})$
20%	3	26	9.8×10^{-5}	$1.4k$	0.987	5.5×10^{-4}	291	$(0.975, 1.1\times10^{-3})$
20%	5	42	9.8×10^{-5}	$1.6k$	0.9999	2.5×10^{-4}	638	$(0.9999, 1.9\times10^{-3})$

last column gives these ratios of the Gilbert and order-6 fits for π_L ($B = 15$), which is designed on the Gilbert fit. Fig. S3 explains the errors through the run persistences of Proposition 1. The Gilbert fit extrapolates the first persistence to all burst depths. It therefore overstates the risk in Rayleigh fading, where persistence first falls with depth, and, for large enough D, understates it under shadowing and on the measured links, where persistence grows with depth. The $M = 2$ class ($\eta = 10^{-4}$) bounds the trace-driven dense risk from above in all 15 configurations and from below in all but the three slow-Rayleigh ones with $D \geq 14$. For π_L ($B = 15$), the trace-driven risk lies below the class range at $f_D = 0.01$ and $D \geq 16$, and above R^+ at $f_D = 0.005$ and $D = 8$.

S5.5 Measured links

Data. We use the FlockLab link-quality dataset [4] (July–September 2019, TelosB, 802.15.4 channel 26, 0 dBm, 8 B payload). In each test, every node sends 100 strobes, and all other nodes record a success/loss bit per strobe. The preprocessed files repeat most (test, sender, receiver) records several times, always with identical content, so records are deduplicated on that key. Rows with a missing or truncated bit string and streams without any success, i.e., with the link down, are discarded. We keep the 171 directed links with loss rate in $[1\%, 25\%]$ and at least 500 distinct tests; these links have 567–805 tests each.

Evaluation. Each test is a short trace; missions of $T = 60$ slide over its 41 possible start positions. Fits and recovery curves are estimated from within-stream counts only. For each link, D is the smallest survival time at which the Gilbert fit certifies $R_{\mathrm{nom}} \leq 10^{-2}$ with dense attempts, and π_L ($B = 15$) is designed on that fit. Standard errors treat tests as independent units; the median relative standard error of the dense risk is 0.29, and differences of more than two standard errors count as significant. Seven links have no failing mission, so their standard error is zero, and the two-standard-error rule counts all of them as significantly below the prediction. By the rule of three over tests, only four of them are, which would change the 20% of significantly lower links in Section V-B to 19%. Fig. S4 shows the whole population, and Table S11 shows the burstiest link of each loss band (1–4, 4–8, 8–15, and 15–25%). The ratios in Table S11 are fitted-to-trace risk ratios; the last column gives this ratio of the Gilbert fit for π_L ($B = 15$) and the measured risk of the recovery-curve redesign. On all four links, the

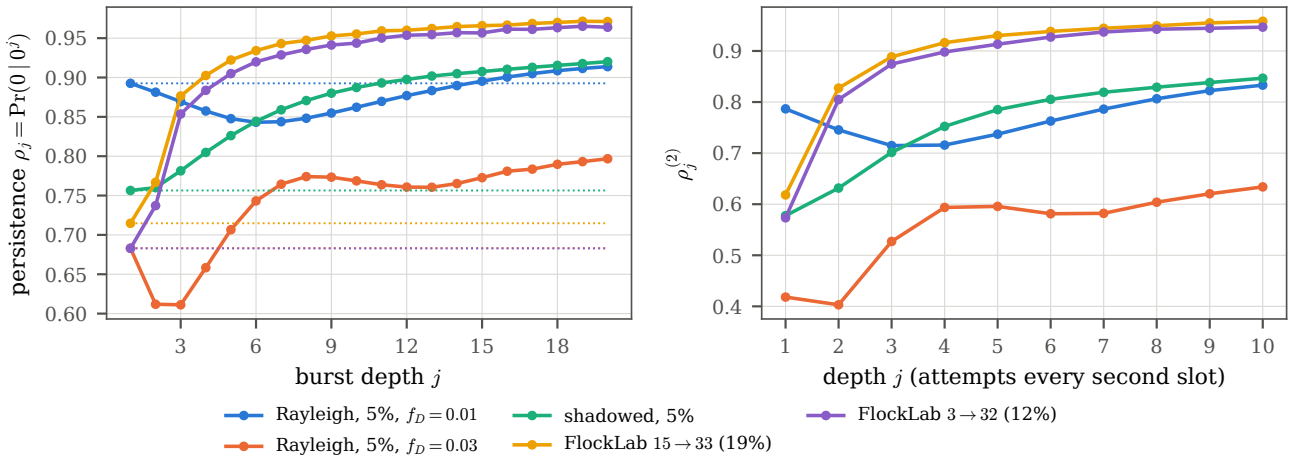


Fig. S3. Run persistences $\rho_j = \Pr(0 \mid 0^j)$ (left) and those of the 2-decimated channel (right) for fading channels and two measured links. Dotted lines show the constant persistence implied by the Gilbert fit.

TABLE S10
ALL FADING CONFIGURATIONS

channel	D	R dense	L=1	2	3	4	6	M=2 class	R, g=2	L=1	R, π_L	L=1/L=6
Rayleigh, 5%, $f_D = 0.01$	12	6.5×10^{-2}	1.3	1.3	1.2	1.2	1.2	$[4.5\times10^{-2}, 9.3\times10^{-2}]$	7.3×10^{-2}	1.3	8.4×10^{-2}	1.2 / 1.1
Rayleigh, 5%, $f_D = 0.01$	16	3.2×10^{-2}	1.6	1.5	1.4	1.2	1.1	$[3.8\times10^{-2}, 7.0\times10^{-2}]$	3.5×10^{-2}	1.6	3.7×10^{-2}	1.6 / 1.1
Rayleigh, 5%, $f_D = 0.01$	20	1.8×10^{-2}	1.7	1.5	1.3	1.1	0.93	$[2.4\times10^{-2}, 5.7\times10^{-2}]$	1.9×10^{-2}	1.7	2.0×10^{-2}	1.7 / 1
Rayleigh, 5%, $f_D = 0.03$	6	8.3×10^{-2}	1.5	1.2	1.2	1.2	1	$[3.4\times10^{-2}, 1.6\times10^{-1}]$	1.1×10^{-1}	1.5	2.1×10^{-1}	1.1 / 1
Rayleigh, 5%, $f_D = 0.03$	8	4.2×10^{-2}	1.4	0.89	0.89	1.1	1.1	$[1.8\times10^{-2}, 1.1\times10^{-1}]$	5.0×10^{-2}	1.5	7.5×10^{-2}	1.5 / 1
Rayleigh, 5%, $f_D = 0.03$	10	2.5×10^{-2}	1.1	0.54	0.54	0.74	0.99	$[5.5\times10^{-3}, 8.7\times10^{-2}]$	2.9×10^{-2}	1.2	3.7×10^{-2}	1.4 / 1.1
Rayleigh, 1%, $f_D = 0.005$	8	2.8×10^{-2}	1	1	1	1	1	$[8.7\times10^{-3}, 2.9\times10^{-2}]$	3.2×10^{-2}	0.96	3.9×10^{-2}	0.88 / 1
Rayleigh, 1%, $f_D = 0.005$	11	1.4×10^{-2}	1.4	1.3	1.2	1.2	1.1	$[8.7\times10^{-3}, 2.0\times10^{-2}]$	1.9×10^{-2}	1.2	1.8×10^{-2}	1.2 / 1.1
Rayleigh, 1%, $f_D = 0.005$	14	7.1×10^{-3}	1.7	1.5	1.3	1.2	1.1	$[8.3\times10^{-3}, 1.5\times10^{-2}]$	7.9×10^{-3}	1.6	9.0×10^{-3}	1.5 / 1.1
shadowed, 5%, $f_D = 0.03$	10	4.9×10^{-2}	1.1	1.1	1.2	1.3	1.3	$[2.5\times10^{-2}, 9.6\times10^{-2}]$	5.4×10^{-2}	1.2	6.4×10^{-2}	1.3 / 1.3
shadowed, 5%, $f_D = 0.03$	14	2.7×10^{-2}	0.6	0.62	0.78	0.94	1.2	$[5.5\times10^{-3}, 6.7\times10^{-2}]$	3.0×10^{-2}	0.66	3.3×10^{-2}	0.76 / 1.2
shadowed, 5%, $f_D = 0.03$	18	1.6×10^{-2}	0.3	0.32	0.45	0.62	0.92	$[1.1\times10^{-3}, 5.3\times10^{-2}]$	1.8×10^{-2}	0.34	1.9×10^{-2}	0.4 / 0.98
shadowed, 1%, $f_D = 0.03$	8	6.6×10^{-3}	0.44	0.71	1	1.2	1.2	$[4.2\times10^{-7}, 1.8\times10^{-2}]$	7.9×10^{-3}	0.56	1.1×10^{-2}	0.82 / 1.1
shadowed, 1%, $f_D = 0.03$	12	2.5×10^{-3}	0.086	0.21	0.46	0.73	1.1	$[5.4\times10^{-11}, 1.1\times10^{-2}]$	2.9×10^{-3}	0.11	3.5×10^{-3}	0.19 / 1.1
shadowed, 1%, $f_D = 0.03$	16	1.0×10^{-3}	0.015	0.053	0.18	0.39	0.81	$[7.0\times10^{-15}, 8.3\times10^{-3}]$	1.2×10^{-3}	0.02	1.4×10^{-3}	0.034 / 0.87

measured dense risk lies within $[R^-, R^+]$ of the $M = 2$ class ($\eta = 10^{-4}$), and the order-6 fit overestimates it by 1.5–2.1×. Over all 171 links, the second-order fit underestimates the dense risk more often than the Gilbert fit, on 85% versus 64% of the links. Part of the growth with burst depth of the measured persistence profiles in Fig. S3, which are pooled over tests, may reflect slow changes in conditions between tests rather than memory within a test; separating the two would require fits per test or per cluster of tests.

TABLE S11
REPRESENTATIVE MEASURED LINKS

link	tests	loss	ℓ	D	R dense	L=1	2	3	4	6	R, π_L	L=1 / rec.
15 → 33	718	18.8%	3.51	18	8.0×10^{-2}	0.1	0.26	1.1	1.3	1.5	1.1×10^{-1}	0.18 / 1.0×10^{-1}
23 → 22	759	7.5%	2.86	13	3.2×10^{-2}	0.23	0.51	1.7	1.8	1.6	4.7×10^{-2}	0.37 / 4.5×10^{-2}
27 → 22	801	2.9%	2.28	9	9.0×10^{-3}	0.73	1.2	3.5	3.2	2.1	1.5×10^{-2}	1.3 / 1.6×10^{-2}
3 → 32	752	11.5%	3.16	15	5.0×10^{-2}	0.17	0.39	1.3	1.6	1.6	7.5×10^{-2}	0.26 / 7.1×10^{-2}

S6. REPRODUCIBILITY

All results except the trace-driven risks and the Monte Carlo check of Table S6 are computed by exact recursions over the joint chain of channel context and scheduler state, in floating-point or exact rational arithmetic. Table S12 maps the claims to scripts in `code/`. The script `verify_paper_claims.py` rechecks the key numerical claims of the main paper and many of this supplement, from the stored results or by direct recomputation, and `make_figs.py` followed by `make_supplement.py` regenerates this document's tables and data figures. Designing π_L for $T = 60$, $B = 15$ takes less than

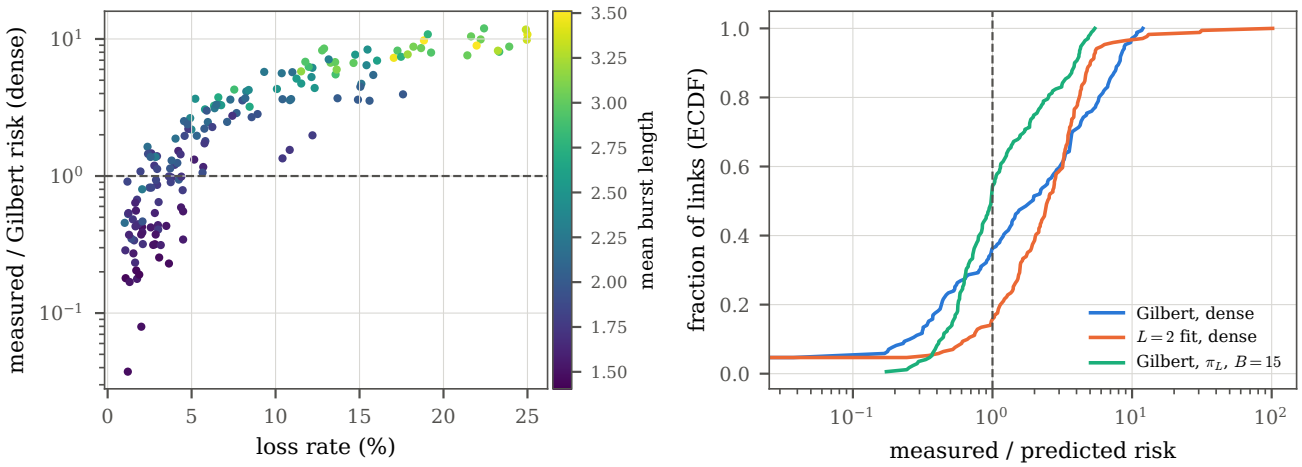


Fig. S4. Measured links (FlockLab, 171 TelosB links). Ratio of measured to Gilbert-predicted dense risk versus loss rate (left), colored by mean burst length, and empirical distribution of measured/predicted risk (right) for the Gilbert fit (dense), the second-order fit (dense), and the Gilbert-predicted risk of π_L ($B = 15$).

a second in exact arithmetic, and evaluating one channel in floating point takes a few milliseconds (Python, one core). The measured traces are downloaded by `get_flocklab.py` (about 50 MB).

TABLE S12
CLAIMS AND THE SCRIPTS THAT PRODUCE THEM

Claim	Script (`code/`)
Lemma 1, Eq. (12), Theorem 2, Fig. S2, Tables S1 and S2	`exp_dense.py`, `exp_dense_M3.py`, `exp_phase.py`
Fig. 2	`make_figs.py`
Proposition 1, decimated persistences	`exp_decimation.py`, `verify_paper_claims.py`
Eq. (14), Proposition S2, reversal criterion	`exp_analytic.py`
Eq. (15), Proposition S3	`exp_closedloop.py`
Proposition 2, exact fractions	`exp_reversal.py`
Fig. 3, Table S3, b-sensitivity, bound tightness	`exp_budget.py`
Certificates, Table S4; η, Table S5; Remark S1	`exp_baselines.py`, `exp_eta.py`, `exp_M3L1.py`
Gilbert–Elliott channels, Table S7	`ge.py`, `exp_ge.py`, `exp_redesign.py`
Redesign, Table S8; redesign on measured links	`exp_redesign.py`, `exp_flocklab.py`
Fig. 4, Table S9	`exp_sweep.py`
Table I (fading), Table S10, Fig. S3	`exp_fading.py`, also with argument `shadow`
Table I (measured), Figs. S3 and S4, Table S11	`get_flocklab.py`, `exp_flocklab.py`
Table S6	`exp_mc_check.py`

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